



NCCER Module 22103



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Identification of Heavy Equipment





Section Two

Vehicle Drive System

Objective

2. Describe the vehicle drive systems used in heavy equipment.
 - a. Describe engines used in heavy equipment.
 - b. Describe auxiliary systems, including fuel, exhaust, cooling, electrical, lubrication, and air filtration systems.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

There are no Performance Tasks associated with this section.



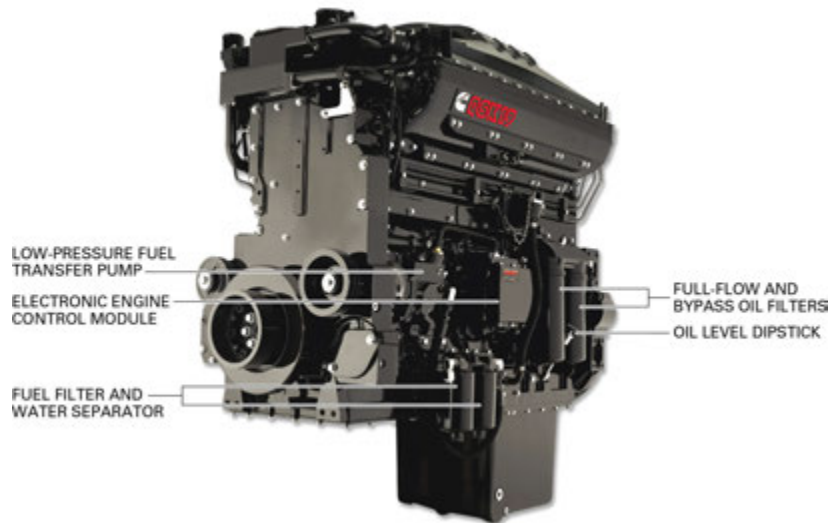
2.0.0 Vehicle Drive System

- The mechanical operation of heavy equipment relies on two separate systems—the vehicle drive system and the hydraulic system.
- Section Two focuses on the vehicle drive system, and the hydraulic system is addressed in Section Three.



2.1.0 Engines (1 of 2)

- Most engines used in construction machines are gasoline-fired or diesel.
- Diesel engines deliver more power to the drive mechanism than gasoline-fired engines.
- In a diesel engine, air and fuel are injected directly into the piston chamber and compressed. The extreme compression creates enough heat to cause the mixture to ignite without a spark plug.



2.1.0 Engines (2 of 2)

- In the United States and Canada, the environmental standards that apply to new diesel engines used in off-road equipment are referred to as Tier-4 standards.
- Tier-4 standards were established to lower the amount of nitrogen oxides (NO_x) and particulate matter (PM) being emitted from engines.
- New diesel engines must be built to burn cleaner ultra-low sulfur diesel fuel.



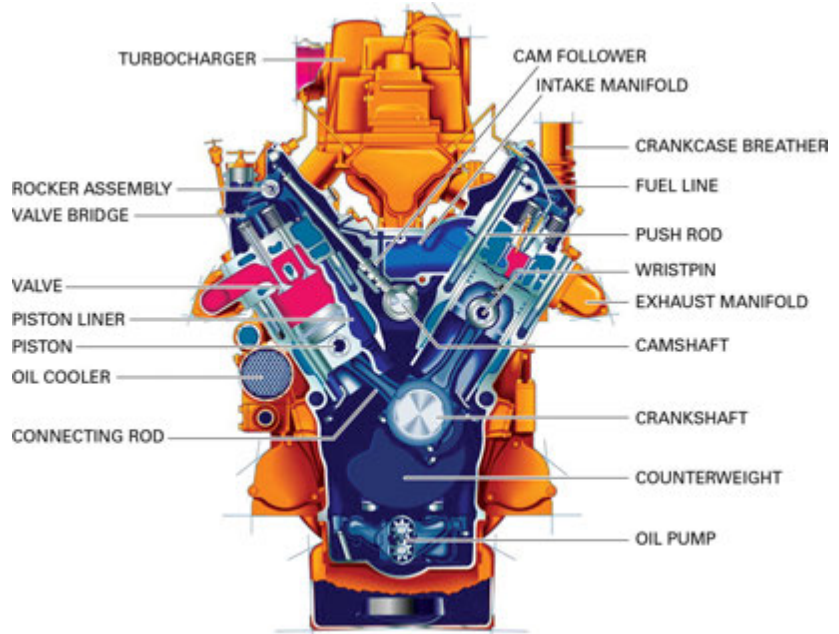
2.2.0 Auxiliary Systems

The engine requires a number of supporting systems, including the fuel, exhaust, cooling, electrical, lubricating, and air filtration systems.



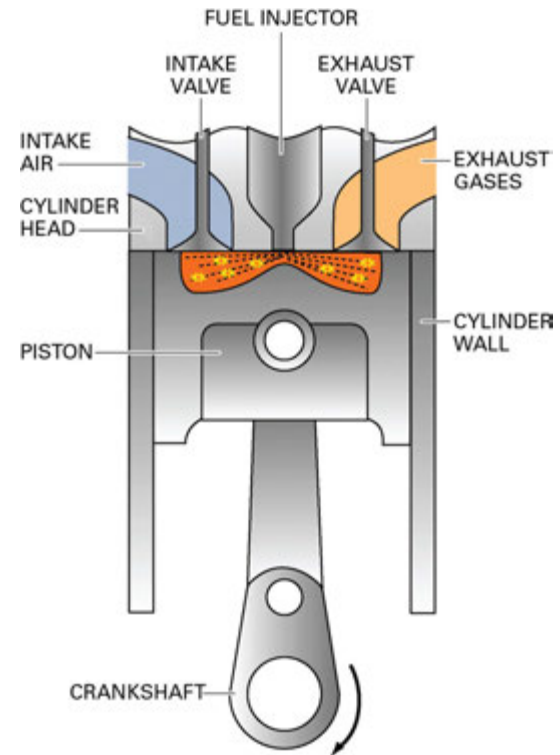
2.2.1 Fuel System (1 of 3)

- The fuel for a diesel engine is injected under high pressure into the combustion cylinder.
- To reduce pollution and increase efficiency, many newer engines control the injection time using electronic controls that monitor a variety of engine functions and conditions during operation.



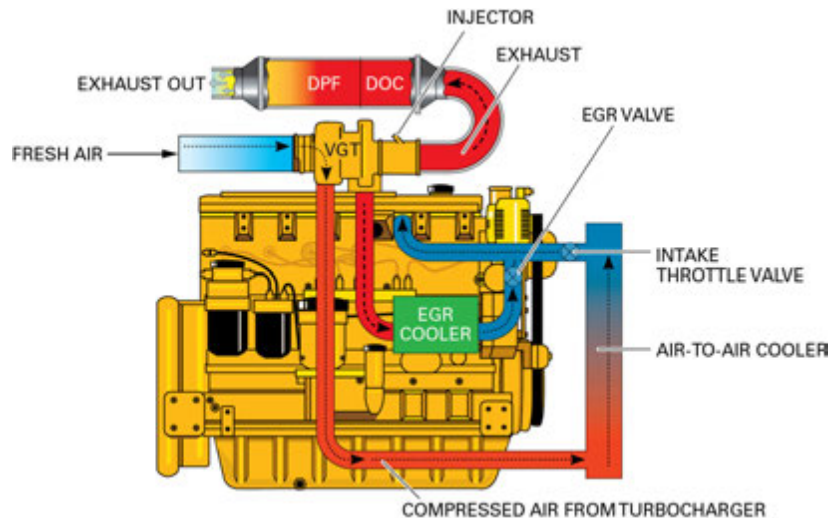
2.2.1 Fuel System (2 of 3)

Turbochargers are designed to increase the amount of air delivered to each of the engine's cylinders, which increases the compression ratio and produces more power.



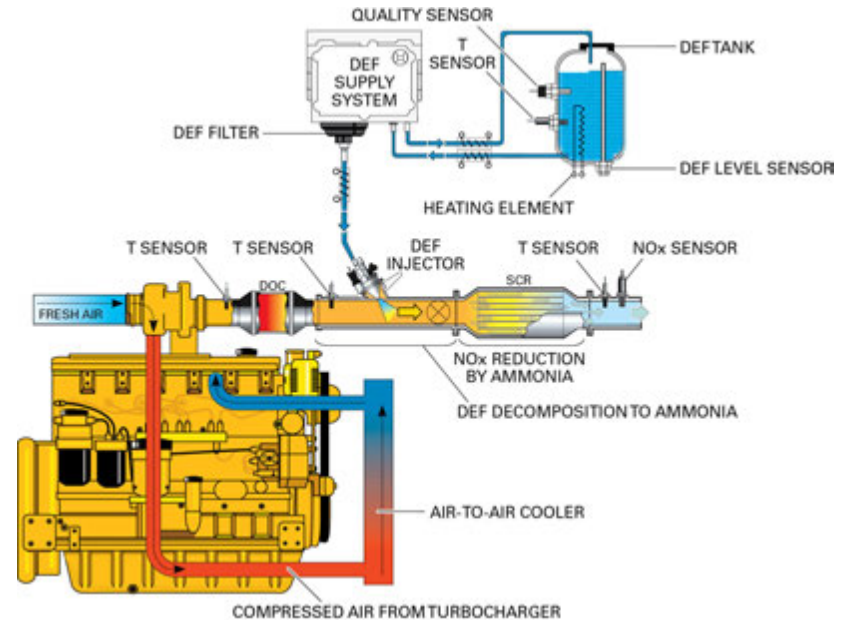
2.2.1 Fuel System (3 of 3)

- One of the ways to reduce the formation of NO_x in an engine is the use of exhaust gas recirculation (EGR) equipment.
- The EGR equipment cools the exhaust gases to lower the temperature of the combustion chamber and reduce the formation of NO_x .



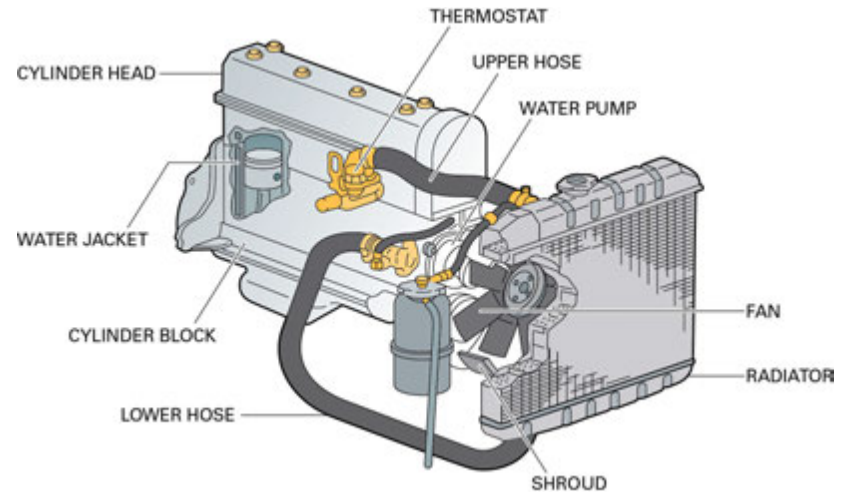
2.2.2 Exhaust System

- The exhaust systems on modern vehicles are designed to minimize the release of hydrocarbons (HC), carbon monoxide, and nitrogen oxides (NO_x).
- Catalytic converters contain materials that chemically affect the gasses passing through the converter to reduce pollutants.
- Selective catalytic reduction (SCR) systems reduce the levels of exhaust NO_x .



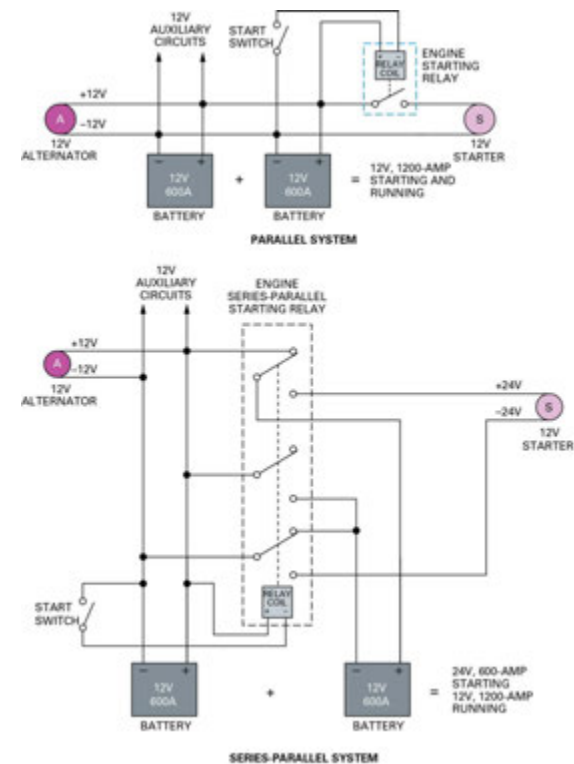
2.2.3 Cooling System

- In a water-cooled system, the engine is kept at the proper temperature by the circulation of pressurized coolant (antifreeze) with a water pump.
- A machine's engine belts, hoses, and coolant level should be checked daily before operation.
- Check the coolant temperature often during operation.
- Never open a hot cooling system.



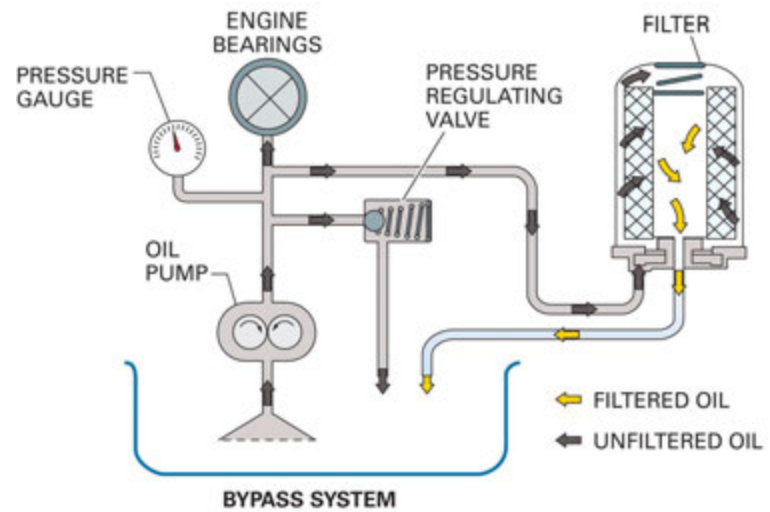
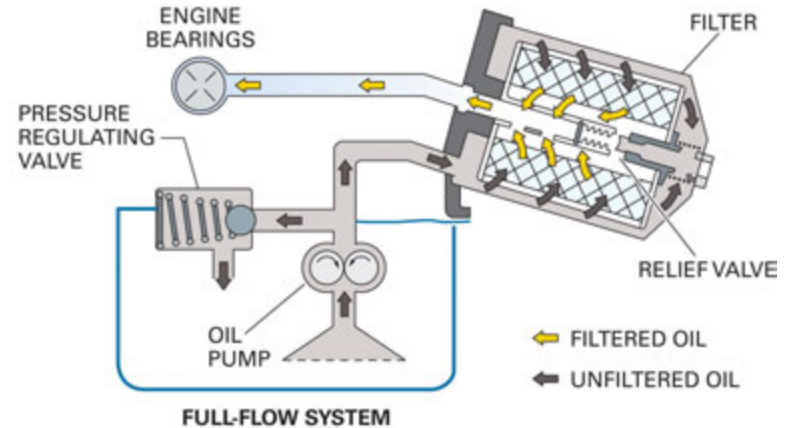
2.2.4 Electrical System

- The typical charging and starting systems consist of a 12V or 24V starting system and an associated 12V **direct current (DC)** alternator.
- Always refer to the manufacturer's operations and maintenance manual before jump-starting.
- Older batteries may give off hydrogen sulfide gas, which can be ignited by sparks.



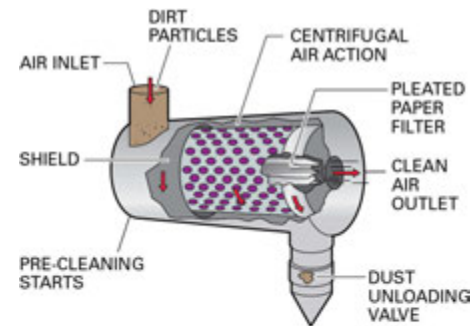
2.2.5 Lubrication System

- This system provides lubrication and cooling to the engine bearings and other friction surfaces internal to the engine.
- When checking the oil in a hot engine, or when changing oil or filters, make sure the engine is stopped.

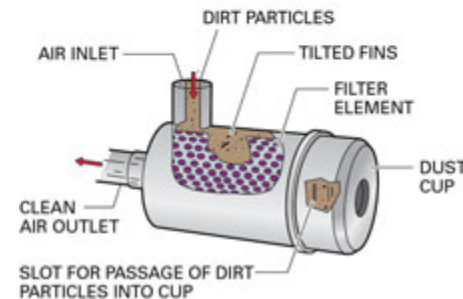


2.2.6 Air Filtration System

- The air intake system to the engine must be monitored by the operator.
- The air is filtered to prevent foreign material, which may cause damage to the engine, from entering the combustion chamber.
- Some filters are equipped with dust cups or dust-unloading valves to rid the filter of some of the accumulated dust.



DRY AIR CLEANER WITH UNLOADING VALVE



DRY AIR CLEANER WITH DUST CUP

Wrap Up – Trade Terms

Direct current (DC): An electrical current that flows in only one direction.



2.0.0 Section Review Question 1

1. The purpose of Tier 4 diesel engine standards is to _____.
- a. increase engine horsepower
 - b. reduce carbon dioxide emissions
 - c. allow engines to burn kerosene
 - d. lower nitrogen oxide emissions



2.0.0 Section Review Question 1 Answer

1. The purpose of Tier 4 diesel engine standards is to _____.
- a. increase engine horsepower
 - b. reduce carbon dioxide emissions
 - c. allow engines to burn kerosene
 - d. lower nitrogen oxide emissions**



2.0.0 Section Review Question 2

2. The air filtration system filters air that is used for _____.
- a. engine cooling
 - b. combustion
 - c. air conditioning
 - d. diluting exhaust



2.0.0 Section Review Question 2 Answer

2. The air filtration system filters air that is used for _____.
- a. engine cooling
 - b. combustion**
 - c. air conditioning
 - d. diluting exhaust



Next...

Section 3.0.0 Hydraulic System



*Read Sections 3.0.0 through 3.2.3 and
complete the Section Review.*